JOL: the layout of one object

Java Object Layout (JOL) reports how the JVM lays an object out: header, fields, padding and size.

```
import org.openjdk.jol.info.ClassLayout;
import org.openjdk.jol.info.GraphLayout;

System.out.println(ClassLayout.parseClass(String.class).toPrintable());

long bytes = GraphLayout.parseInstance(list).totalSize();   // the object and all it references
```

- `ClassLayout` shows one object; `GraphLayout` adds up everything reachable from it.
- JOL is an OpenJDK project, added as a library; it is not part of the JDK.

JOL with compact object headers

`HashMap$Node`	Header	Fields	Size
Default	12 B	hash, key, value, next: 16 B	32 B (4 B padding)
`-XX:+UseCompactObjectHeaders`	8 B	the same 16 B	24 B

- Run the same JOL code with the flag off and on, and compare the sizes.
- The 4 B saved only counts when the total drops below the next 8 B boundary.

References

- OpenJDK Code Tools: Java Object Layout (JOL)
- The `java` command (Java SE 25): `-XX:+UseCompactObjectHeaders`